

Dunnett's Critical Value Table

Two-Tailed Critical Values

Significance Levels: $\alpha = 0.05$ and $\alpha = 0.01$

k = 2 to 9 treatment groups | Error df = 2 to ∞

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How to use this table: After running a one-way ANOVA with a significant F-test, identify (1) k — the number of treatment groups excluding the control, and (2) df_error from your ANOVA table (N - k - 1). Find the row for df_error and the column for k. The intersection is d*. For each treatment i, compute $d = (\text{mean}_i - \text{mean}_0) / \text{SE}$ where $\text{SE} = \sqrt{\text{MSE} * (1/n_i + 1/n_0)}$. Reject H0 if $|d| > d^*$.

Use two-tailed tables when treatments could differ from control in either direction (higher or lower). This is the standard choice for most experiments.

Dunnett's Two-Tailed Critical Values | alpha = 0.05 (95% Confidence)

Reject H0 for treatment i if $|d| > d^*$ (two-tailed) | Yellow row = df = 20 (most common in practice)

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	2.57	3.03	3.29	3.48	3.62	3.73	3.82	3.90
3	2.34	2.68	2.88	3.04	3.16	3.26	3.34	3.41
4	2.25	2.56	2.74	2.88	3.00	3.09	3.17	3.24
5	2.20	2.49	2.67	2.80	2.91	3.00	3.07	3.14
6	2.17	2.45	2.62	2.75	2.86	2.94	3.01	3.08
7	2.15	2.43	2.59	2.72	2.82	2.91	2.97	3.04
8	2.13	2.41	2.57	2.70	2.79	2.88	2.94	3.01
9	2.12	2.40	2.55	2.68	2.77	2.86	2.92	2.99
10	2.11	2.38	2.54	2.66	2.76	2.84	2.91	2.97
11	2.10	2.37	2.53	2.65	2.74	2.83	2.89	2.96

df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
12	2.10	2.37	2.52	2.64	2.73	2.82	2.88	2.95
13	2.09	2.36	2.51	2.63	2.73	2.81	2.87	2.94
14	2.09	2.36	2.51	2.63	2.72	2.80	2.86	2.93
15	2.09	2.35	2.50	2.62	2.72	2.80	2.86	2.92
16	2.08	2.35	2.50	2.62	2.71	2.79	2.85	2.92
17	2.08	2.35	2.50	2.61	2.71	2.79	2.85	2.91
18	2.08	2.34	2.49	2.61	2.70	2.78	2.84	2.91
19	2.08	2.34	2.49	2.61	2.70	2.78	2.84	2.91
20	2.08	2.34	2.49	2.61	2.70	2.78	2.84	2.90
24	2.06	2.33	2.47	2.59	2.68	2.76	2.82	2.88
30	2.05	2.31	2.46	2.57	2.66	2.74	2.80	2.86
40	2.04	2.30	2.44	2.56	2.64	2.72	2.78	2.84
60	2.02	2.28	2.42	2.53	2.62	2.70	2.76	2.81
120	2.01	2.27	2.41	2.52	2.61	2.68	2.74	2.80
∞	2.00	2.26	2.40	2.51	2.59	2.67	2.73	2.78

Source: Dunnett (1955), JASA 50:1096-1121. Values agree with Kutner et al. (2005), Appendix B, Table B.9.

Test Statistic

$$d = (\text{mean}_i - \text{mean}_0) / SE$$

Standard Error

$$SE = \sqrt{MSE \times (1/n_i + 1/n_0)}$$

Error Degrees of Freedom

$$df_{\text{error}} = N - (k + 1)$$

Decision Rule

$$\text{Reject } H_0 \text{ if } |d| > d^*$$

Dunnett's Two-Tailed Critical Values $\alpha = 0.01$ (99% Confidence) Use for strict significance testing where Type I error control is critical Yellow row = df = 20								
df \ k	k = 2	k = 3	k = 4	k = 5	k = 6	k = 7	k = 8	k = 9
2	3.70	4.15	4.43	4.63	4.79	4.92	5.03	5.12
3	3.09	3.43	3.64	3.80	3.93	4.03	4.12	4.20
4	2.88	3.17	3.35	3.50	3.62	3.71	3.79	3.87
5	2.76	3.03	3.20	3.33	3.44	3.53	3.61	3.68
6	2.70	2.95	3.11	3.24	3.34	3.43	3.50	3.57
7	2.65	2.90	3.05	3.18	3.27	3.36	3.43	3.50
8	2.62	2.86	3.01	3.13	3.23	3.31	3.38	3.45
9	2.59	2.83	2.98	3.10	3.19	3.28	3.35	3.41
10	2.57	2.81	2.95	3.07	3.16	3.24	3.32	3.38
11	2.56	2.79	2.93	3.05	3.14	3.22	3.29	3.36
12	2.55	2.78	2.92	3.03	3.13	3.21	3.28	3.34
13	2.54	2.77	2.91	3.02	3.11	3.19	3.26	3.33
14	2.53	2.76	2.90	3.01	3.10	3.18	3.25	3.31
15	2.53	2.75	2.89	3.00	3.09	3.17	3.24	3.30
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120	2.44	2.64	2.77	2.87	2.95	3.02	3.09	3.15
∞	2.42	2.63	2.75	2.85	2.93	3.00	3.07	3.12

Source: Dunnett (1955), JASA 50:1096-1121. Validated against R DunnettTest() and NIST/SEMATECH e-Handbook.

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References & Sources

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- [1] Dunnett, C.W. (1955). A multiple comparison procedure for comparing several treatments with a control. *Journal of the American Statistical Association*, 50(272), 1096-1121. <https://doi.org/10.2307/2281208>
- [2] Kutner, M.H., Nachtsheim, C.J., Neter, J., & Li, W. (2005). *Applied Linear Statistical Models* (5th ed.). McGraw-Hill. Chapter 17.
- [3] NIST/SEMATECH (2012). *e-Handbook of Statistical Methods* — Multiple Comparisons. <https://www.itl.nist.gov/div898/handbook/prc/section4/prc471.htm>
- [4] Penn State STAT 503: Design of Experiments — Multiple Comparisons. <https://online.stat.psu.edu/stat503/lesson/3/3.3>
- [5] U.S. Food & Drug Administration. (2022). *Multiple Endpoints in Clinical Trials: Guidance for Industry*. FDA. <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/multiple-endpoints-clinical-trials-guidance-industry>